Chapter 3. Permutation Tests

Randomization and population models, conditional inference, permutation tests for common problems

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3.1 Two sources of randomness

- **Randomization model.** N fixed units are randomly assigned, m to treatment 1 and $n = N - m$ to treatment 2. The responses are fixed numbers; only the assignment is random. Inference concerns these N units.
- **Population model.** $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are random samples. Inference concerns F and G, e.g. $H_0 : F = G$.

Both lead to the same computation: the *permutation* (or *randomization*) distribution of a statistic over all re-assignments of the observed values.

3.2 The randomization test

Under H_0: "treatment has no effect," each unit's response would be the same under either treatment. The observed assignment is one of $\binom{N}{m}$ equally likely assignments, and every other assignment would have produced the same pooled set of responses. For a statistic D (e.g. $\bar{Y} - \bar{X}$), let $d_1, \dots, d_K$, $K = \binom{N}{m}$, be its values over all assignments. The upper-tailed p-value is

$$p = \frac{1}{K} \sum_{k=1}^{K} I(d_k \geq D_{\text{obs}}). \tag{3.1}$$

For a two-sided test use $|d_k| \geq |D_{\text{obs}}|$ (or a symmetric criterion), not twice the smaller one-sided value: the randomization distribution need not be symmetric.

Example 3.1

Five patients: two random patients get a new treatment with recovery times $3, 5$; the others have $8, 9, 12$. Let $D = \bar{Y}_{\text{old}} - \bar{X}_{\text{new}}$. The pooled total is 37, so for a treated pair with sum s, $D = (37 - s)/3 - s/2$, a decreasing function of s. The $\binom{5}{2} = 10$ pair sums are $8, 11, 12, 13, 14, 15, 17, 17, 20, 21$. The observed sum 8 is the smallest, so $D_{\text{obs}} = 5.67$ is the largest value and $p = 1/10$.

Proposition 3.1 (Exactness)

Under H_0, if the d_k have no ties and $\alpha = j/K$, the randomization test rejecting when $p \leq \alpha$ has type I error exactly α. With ties it is at most α.

Proof. Under H_0 the multiset $\{d_k\}$ does not depend on which assignment occurred, and the realized assignment is uniform. So D_{obs} is a uniformly chosen element of the multiset. Without ties, $p \leq j/K$ iff D_{obs} is among the j largest values, which has probability j/K. With ties, p only gets larger. ■

Proposition 3.2 (Equivalent statistics)

Over all re-assignments of a fixed pooled sample, $\sum_j Y_j$, $\bar{Y} - \bar{X}$ and the pooled two-sample t-statistic are strictly increasing functions of each other, hence give identical permutation p-values.

Proof. The pooled total $\sum X_i + \sum Y_j$ is fixed, so $\bar{Y} - \bar{X} = \frac{N}{mn} \sum Y_j - \frac{\text{total}}{m}$ is increasing in $\sum Y_j$. The total sum of squares SST is also fixed, and $SST = (N-2)s_p^2 + \frac{mn}{N} D^2$ with $D = \bar{Y} - \bar{X}$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m} + \frac{1}{n}} \sqrt{(SST - \frac{mn}{N} D^2)/(N-2)}} = g(D), \qquad g'(D) \propto \frac{SST}{(SST - \frac{mn}{N} D^2)^{3/2}} > 0.$$

■

Statistics such as a difference of medians or of trimmed means are *not* monotone in D and can give different (often more powerful, for skewed or heavy-tailed data) tests.

3.3 Permutation tests in the population model

Definition (Exchangeability)

$(Z_1, \dots, Z_N)$ is **exchangeable** if $(Z_{\pi(1)}, \dots, Z_{\pi(N)}) \overset{d}{=} (Z_1, \dots, Z_N)$ for every permutation π.

I.i.d. variables are exchangeable. So are equicorrelated normal variables $N_N(\mu\mathbf{1}, \sigma^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}'])$, which are not independent. Two independent samples from $N(\mu, 1)$ and $N(\mu, 4)$ are *not* exchangeable, even though their means agree.

Theorem 3.3 (Validity of permutation tests)

Let $Z = (Z_1, \dots, Z_N)$ be exchangeable under H_0, let $\mathcal{G}$ be a group of permutations, and let $p(Z) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} I(T(gZ) \geq T(Z))$. Then $P_{H_0}(p(Z) \leq \alpha) \leq \alpha$ for every α, with equality when $\alpha|\mathcal{G}|$ is an integer and there are no ties. The statement holds conditionally on the multiset of observed values, hence also unconditionally.

Proof. Let U be uniform on $\mathcal{G}$, independent of Z. By exchangeability $UZ \overset{d}{=} Z$. Since $\mathcal{G}$ is a group, the orbit $\{gUZ : g \in \mathcal{G}\}$ equals $\{gZ\}$, so $p(UZ)$ is computed from the same multiset of values $\{T(gZ)\}$, and $T(UZ)$ is a uniformly chosen member of it. A uniformly chosen member lies among the top j (counting ties generously) with probability at most $j/|\mathcal{G}|$. Hence $P(p(Z) \leq \alpha) = P(p(UZ) \leq \alpha) \leq \alpha$. Taking expectations over the conditioning gives the unconditional statement. ■

For two samples under $H_0 : F = G$ the pooled observations are i.i.d., hence exchangeable, and the relevant re-assignments are the $\binom{N}{m}$ splits (each realized by $m!\,n!$ permutations). The permutation test is **conditional inference**: the observed values are held fixed and only their labels are random.

3.4 Random (Monte Carlo) permutation tests

When $\binom{N}{m}$ is too large, draw M random relabelings, compute $T_1, \dots, T_M$, set $T_0 = T_{\text{obs}}$, and use

$$\hat{p} = \frac{1 + \sum_{i=1}^{M} I(T_i \geq T_{\text{obs}})}{M + 1}. \tag{3.2}$$

Proposition 3.4

Under H_0, $P(\hat{p} \leq \alpha) \leq \alpha$ for every M. Moreover $\hat{p}$ estimates the exact permutation p-value with standard error about $\sqrt{p(1-p)/M}$.

Proof. Given the multiset of values, $T_0, T_1, \dots, T_M$ are exchangeable under H_0: the observed labeling is itself a uniform draw, like the M random ones. The rank of T_0 among the $M+1$ values is therefore uniform (with ties broken against rejection), and $\hat{p} \leq j/(M+1)$ iff T_0 is among the j largest. For the second claim, $\hat{p} \approx$ the mean of M Bernoulli(p) indicators. ■

This is why the "+1" is used: it makes the Monte Carlo test exactly valid, not just approximately.

Generating a random relabeling. Draw a uniform random permutation of $1, \ldots, N$, and give group 1 either to the units in its first m positions or to the units that receive the first m labels. Either way, each m-subset arises from exactly $m!\,n!$ of the $N!$ permutations, so every split has probability $1/\binom{N}{m}$.

Two-sample alternatives and p-values. With $H_0 : F = G$, the one-sided alternatives are stochastic orderings, $G(x) \le F(x)$ for all x (Y tends to be larger) or the reverse. Under a shift model they become $\Delta > 0$ or $\Delta < 0$. For $D = \bar{Y} - \bar{X}$ with values $d_0 = D_{\text{obs}}, d_1, \ldots, d_M$:

$$p_{\text{upper}} = \frac{\#\{d_i \ge d_0\}}{M+1}, \qquad p_{\text{lower}} = \frac{\#\{d_i \le d_0\}}{M+1}, \qquad p_{\text{two}} = \frac{\#\{|d_i| \ge |d_0|\}}{M+1}.$$

With $M = 1999$, the standard error at $p = 0.05$ is about 0.005.

3.5 Other permutation tests

One sample: sign flipping. If $X_i - \theta_0$ are independent and symmetric about 0 under H_0, then $(\pm(X_1 - \theta_0), \ldots, \pm(X_n - \theta_0))$ have the same distribution for all 2^n sign choices. The permutation test over sign flips with statistic $\sum I(X_i > \theta_0)$ is the sign test. With $\sum V_i R_i$ it is the signed-rank test. With the one-sample t-statistic it is a symmetric-F version of the t-test (so the tests of Chapter 2 are permutation tests).

2×2 tables: Fisher's exact test. With two binary variables (e.g. exposure and disease) and H_0: independence (or equal proportions), hold one variable fixed and permute the other. With margins fixed, the count D in the (exposed, diseased) cell follows the hypergeometric distribution:

$$P(D = d) = \binom{m}{d}\binom{N-m}{k-d} \Big/ \binom{N}{k},$$

where m is the number diseased and k the number exposed among N.

Example 3.2

$N = 20$ subjects, $m = 10$ diseased, $k = 6$ exposed, and 5 of the exposed are diseased. Then $P(D \ge 5) = [\binom{10}{5}\binom{10}{1} + \binom{10}{6}\binom{10}{0}]/\binom{20}{6} = (2520 + 210)/38760 = 0.070$.

Summary. Permutation tests (i) need only exchangeability under H_0, (ii) allow any statistic, and (iii) give exact p-values by computation. Their limits: they test only hypotheses that make the data exchangeable (e.g. not "equal means, unequal variances"), and interval estimation requires inverting many tests.